

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

Video ID:	be1a9e91-40ad-4596-9a92-9d3e4080cc86
Total Frames:	3006
Frame Rate:	25.0 fps (assumed)
Duration:	120.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	113	✓
RT-DETR Detections	1611	✓
MedSAM2 Detections	1514	✓
Consensus (3-Model Agreement)	17	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	1	5.9%
LOW RISK	3	17.6%
TOTAL ANALYZED	17	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 17 frames)



Feature	Value	Interpretation
Frame Index	244	Frame 244 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 61.5%
Redness Score	0.002	Color-based vascularization
Relative Radius	8.8%	Polyp size as % of frame diagonal
Texture Score	0.305	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	71.7%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 141 frames)

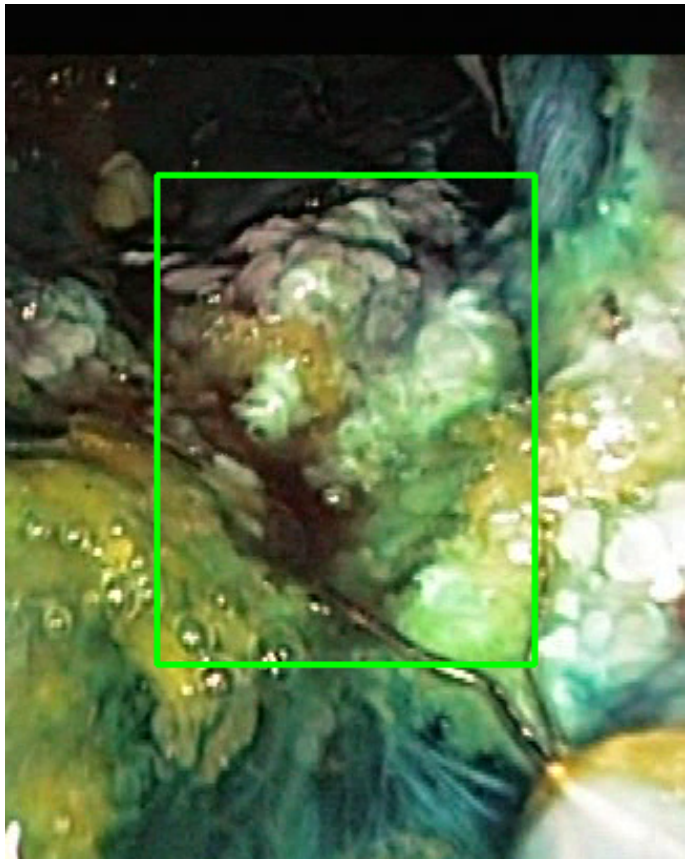


Feature	Value	Interpretation
Frame Index	2725	Frame 2725 of 3006
Detection Method	Detector Consensus (MedSAM2 / RT-DETR)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 62.4%
Redness Score	0.003	Color-based vascularization
Relative Radius	37.5%	Polyp size as % of frame diagonal
Texture Score	0.155	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	71.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 128 frames)



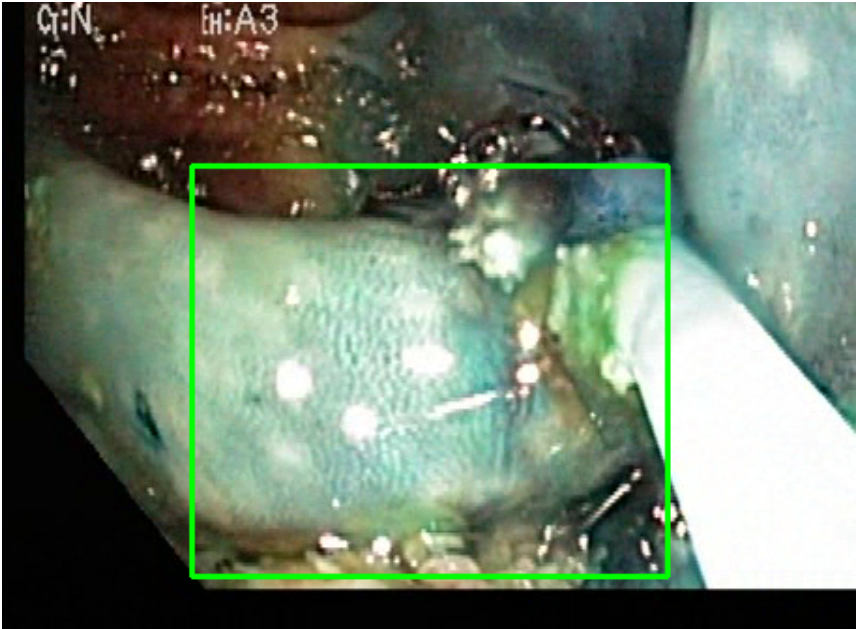
Feature	Value	Interpretation
Frame Index	1728	Frame 1728 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 70.6%
Redness Score	0.012	Color-based vascularization
Relative Radius	38.2%	Polyp size as % of frame diagonal
Texture Score	0.250	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	65.0%	YOLO / RT-DETR / track confidence
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Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — LIFTED_POLYP [LOW_RISK] (Tracked across 24 frames)

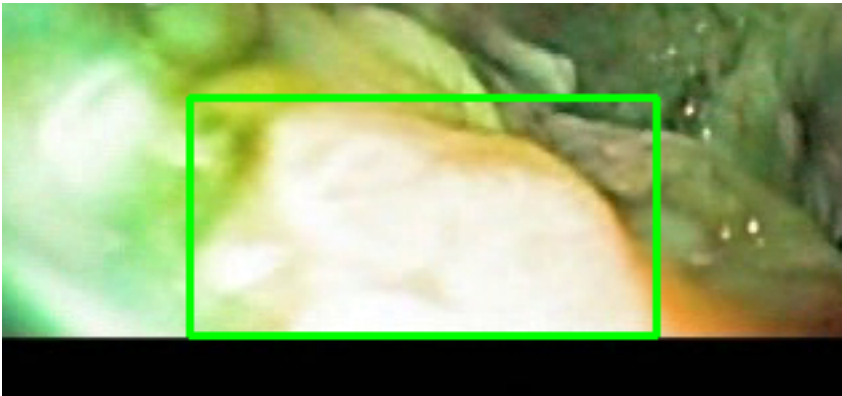


Feature	Value	Interpretation
Frame Index	1918	Frame 1918 of 3006
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 73.5%
Redness Score	0.003	Color-based vascularization
Relative Radius	5.6%	Polyp size as % of frame diagonal
Texture Score	0.482	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	64.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #5 — LIFTED_POLYP [LOW_RISK] (Tracked across 168 frames)



Feature	Value	Interpretation
Frame Index	556	Frame 556 of 3006
Detection Method	Detector Consensus (YOLO / RT-DETR)	Type confidence: 70.6%
POLYP TYPE	LIFTED_POLYP	Type confidence: 70.6%
Redness Score	0.016	Color-based vascularization
Relative Radius	19.6%	Polyp size as % of frame diagonal
Texture Score	0.365	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	69.8%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #6 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 17 frames)



Feature	Value	Interpretation
Frame Index	664	Frame 664 of 3006
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 85.5%
Redness Score	0.159	Color-based vascularization
Relative Radius	12.8%	Polyp size as % of frame diagonal
Texture Score	0.173	Surface morphology
Vessel Visibility	0.122	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	64.9%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #7 — LIFTED_POLYP [LOW_RISK] (Tracked across 39 frames)



Feature	Value	Interpretation
Frame Index	2620	Frame 2620 of 3006
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 77.9%
Redness Score	0.056	Color-based vascularization
Relative Radius	19.4%	Polyp size as % of frame diagonal
Texture Score	0.334	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	85.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #8 — LIFTED_POLYP [LOW_RISK] (Tracked across 40 frames)

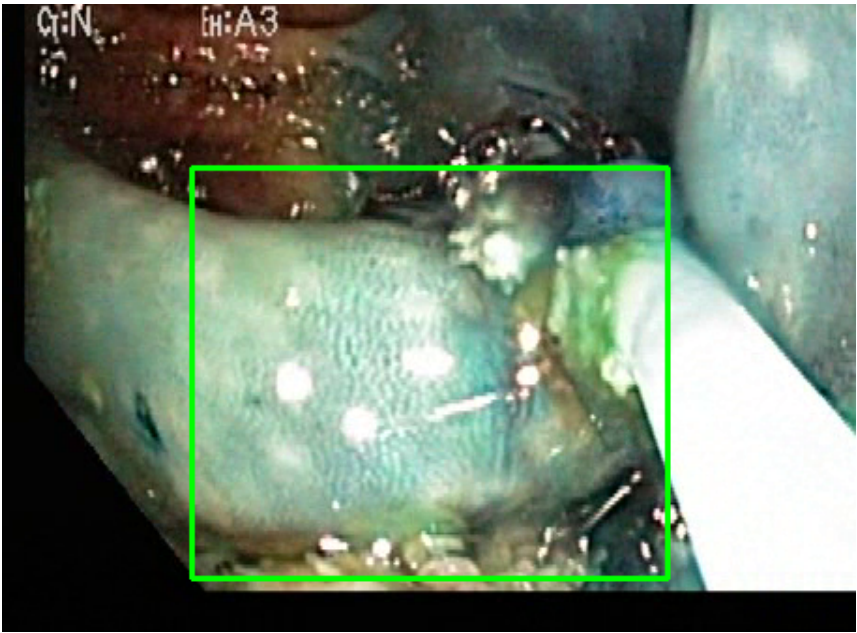


Feature	Value	Interpretation
Frame Index	114	Frame 114 of 3006
Detection Method	Detector Consensus (YOLO SAM2 DETR)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 91.6%
Redness Score	0.000	Color-based vascularization
Relative Radius	12.3%	Polyp size as % of frame diagonal
Texture Score	0.720	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	70.6%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #9 — LIFTED_POLYP [LOW_RISK] (Tracked across 24 frames)



Feature	Value	Interpretation
Frame Index	1918	Frame 1918 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 73.5%
Redness Score	0.003	Color-based vascularization
Relative Radius	5.6%	Polyp size as % of frame diagonal
Texture Score	0.482	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	66.5%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #10 — LIFTED_POLYP [LOW_RISK] (Tracked across 23 frames)



Feature	Value	Interpretation
Frame Index	2952	Frame 2952 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 84.7%
Redness Score	0.011	Color-based vascularization
Relative Radius	7.1%	Polyp size as % of frame diagonal
Texture Score	0.659	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	76.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #11 — LIFTED_POLYP [LOW_RISK] (Tracked across 39 frames)



Feature	Value	Interpretation
Frame Index	2620	Frame 2620 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 77.9%
Redness Score	0.056	Color-based vascularization
Relative Radius	19.4%	Polyp size as % of frame diagonal
Texture Score	0.334	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	66.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #12 — LIFTED_POLYP [LOW_RISK] (Tracked across 19 frames)

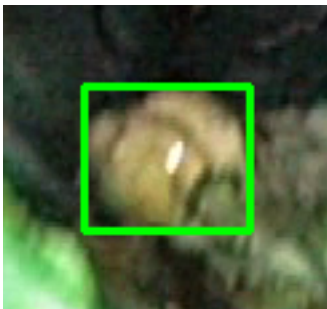


Feature	Value	Interpretation
Frame Index	2928	Frame 2928 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 76.3%
Redness Score	0.015	Color-based vascularization
Relative Radius	11.4%	Polyp size as % of frame diagonal
Texture Score	0.457	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	74.7%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #13 — LIFTED_POLYP [LOW_RISK] (Tracked across 19 frames)

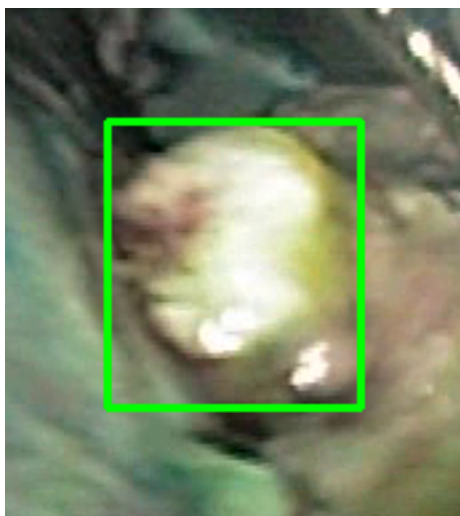


Feature	Value	Interpretation
Frame Index	2546	Frame 2546 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 66.6%
Redness Score	0.004	Color-based vascularization
Relative Radius	34.1%	Polyp size as % of frame diagonal
Texture Score	0.157	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	70.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #14 — LIFTED_POLYP [LOW_RISK] (Tracked across 19 frames)

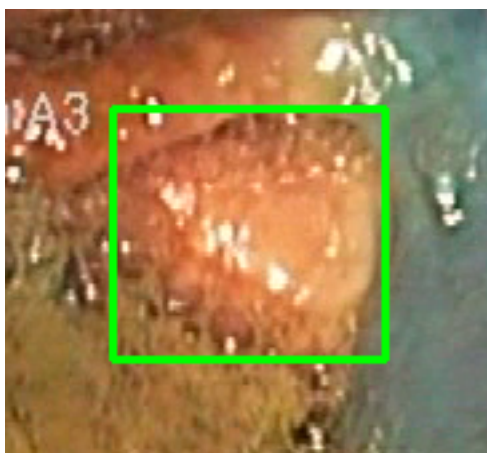


Feature	Value	Interpretation
Frame Index	631	Frame 631 of 3006
Detection Method	Detector Consensus (MedSAM2 / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 59.3%
Redness Score	0.009	Color-based vascularization
Relative Radius	5.7%	Polyp size as % of frame diagonal
Texture Score	0.147	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	69.3%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #15 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 18 frames)



Feature	Value	Interpretation
Frame Index	671	Frame 671 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 92.0%
Redness Score	0.167	Color-based vascularization
Relative Radius	11.8%	Polyp size as % of frame diagonal
Texture Score	0.349	Surface morphology
Vessel Visibility	0.128	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	78.3%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #16 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 17 frames)

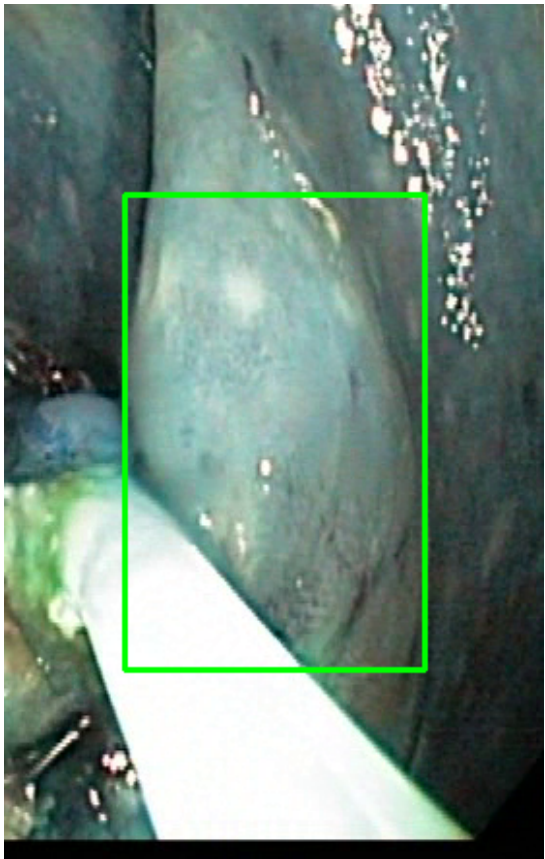


Feature	Value	Interpretation
Frame Index	664	Frame 664 of 3006
Detection Method	Detector Consensus (MedSAM2 / DETR)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 85.5%
Redness Score	0.159	Color-based vascularization
Relative Radius	12.8%	Polyp size as % of frame diagonal
Texture Score	0.173	Surface morphology
Vessel Visibility	0.122	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	78.7%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #17 — LIFTED_POLYP [LOW_RISK] (Tracked across 24 frames)



Feature	Value	Interpretation
Frame Index	1918	Frame 1918 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 73.5%
Redness Score	0.003	Color-based vascularization
Relative Radius	5.6%	Polyp size as % of frame diagonal
Texture Score	0.482	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	72.1%	YOLO / RT-DETR / track confidence
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Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:

A total of 17 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 1 (5.9%)
- LOW RISK polyps: 3 (17.6%)

Recommendation:

The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.